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Hamid Ettayyebi

Data Scientist | AI Engineer

Portfolio: hamid701.github.io
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PROFESSIONAL SUMMARY

Data Scientist and AI Engineer with research-to-production experience in time series, machine learning, and RAG systems. Combines a strong foundation in mathematics and statistics with reproducible, best-practice engineering, taking solutions from modeling through production deployment. Delivers end-to-end, production-grade solutions for stakeholders.

SKILLS

- **Programming & Data:** Python, SQL, Pandas, NumPy, SciPy, PostgreSQL, SQLAlchemy.
- **Data Science & ML:** Time-series, statistical modeling, anomaly detection, scikit-learn, XGBoost, TensorFlow, Keras.
- **AI Engineering:** RAG, LLM applications, GenAI, embeddings, semantic search, pgvector, LangChain, LLM evaluation.
- **Engineering & Deployment:** FastAPI, Docker, Google Cloud (Cloud Run), Git, GitHub, CI/CD, pytest.
- **Visualization:** Plotly, Dash, Streamlit, Matplotlib, Seaborn, GeoPandas, Folium.
- **Languages:** Arabic (native), English (C1), French (B2), Italian (A2).

PROFESSIONAL EXPERIENCE

Data Scientist **Jan 2026–present**
Hermes Corporate | Client: Baker Hughes *Florence, Italy*

- Built and deployed end-to-end gas-turbine mineral-oil leak detector using wavelet denoising and multi-window regression on multiple sensor streams, reducing detection time from 13 hours to under 2 hours. Deployment across a 1,000-turbine fleet carried a projected benefit of over €1M per year in avoided downtime and equipment damage.
- Integrated adaptive per-asset thresholds, PostgreSQL-backed historical statistics, and cross-batch state handling into the pipeline; supported by a 255-test regression suite to protect detection behavior during changes and support reliable production operation.
- Developed and deployed end-to-end crank-performance detection with phase-based signal analysis and historical-deviation methods, including PostgreSQL storage and asset-specific history queries, to flag abnormal starts and help engineers prioritize investigation before potential start failures.

PhD Researcher: Applied Data Scientist **Jan 2017–Sep 2021**
Mohammed V University *Rabat, Morocco*

- Developed neural-network and statistical models for solar-irradiance forecasting, with an ANN model outperforming statistical baselines by 13.43%, supporting more accurate estimates of solar availability for energy planning.
- Built automated data-preparation and model-validation workflows that reduced preparation time by 30%, enabling faster comparison of forecasting approaches and more reproducible research.

Mathematics Teacher **Sep 2017–Aug 2024**
Regional Academy for Training and Education *Rabat, Morocco*

- Taught more than 1,120 students and trained five peer tutors, helping raise the pass rate from 82% to 92% through structured explanations, practice, and targeted support. success by 25%

PERSONAL PORTFOLIO PROJECT

Turbofan Maintenance Copilot **July 2026–Oct 2026**
End-to-end project - GitHub | Live application (access key required) *Rome, Italy*

- Built and deployed a RAG assistant for FAA maintenance manuals using BGE embeddings, PostgreSQL/pgvector, and hybrid retrieval, improving first-result retrieval success from 56% to 78% on nine labeled questions; added page citations and unsupported-question checks, and shipped with FastAPI, Docker, Google Cloud Run, CI/CD, and 240+ automated tests.
- Developed a gradient-boosted remaining-useful-life model on NASA C-MAPSS sensor data for predictive maintenance, reducing MAE from 25.8 to 8.4 cycles against the degradation-index baseline on 100 held-out engines.

Rentelligence AI: Dynamic Pricing Engine **Jan 2025–Apr 2025**
End-to-end project - GitHub | Live demo *Rome, Italy*

- Built an XGBoost model and Streamlit application using more than 12,000 rental listings, data cleaning, property-feature engineering, and geographic analysis; achieved test-set MAE of €304 and R^2 of 0.74, giving renters and property owners a data-based reference for comparing asking prices.

EDUCATION & CERTIFICATIONS

Master's Degree in Mathematics and Applications, Statistics, and Numerical Computing
Mohammed V University *Rabat, Morocco*

IBM Data Science Specialization **Feb 2025**
Credential ID: W85E3XU7YR5X